Case Study Challenge Brief 2026: Sustainable Urban Mobility Infrastructure. Teams are required to evaluate the capital deployment feasibility and regulatory constraints of transitioning municipal transit networks to zero-emission battery electric bus fleets across Tier-1 metropolitan areas. Address total cost of ownership including depot charging infrastructure, grid interconnect upgrade timelines, lithium iron phosphate battery lifecycle degradation under extreme ambient thermal conditions, and route schedule optimization. Submissions must provide rigorous financial models and operational risk mitigation frameworks for municipal transport authorities.

Grid Integration and Depot Solar Microgrid Peak-Shaving:

Co-locating two-megawatt rooftop solar photovoltaic arrays with containerized lithium battery stationary storage at municipal bus depots cushions localized substation transformers from sudden transformer thermal overloading during synchronous late-night depot recharging.